First Density Plots Results Multiverse

2026-04-21

SETUP

```r
# Load required packages
library(here)
```

```
here() starts at /Users/loesjeubbink/Desktop/MSBBSS uu/Jaar 2/Many Analysist Thesis/ManyAnaly
```

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(ggplot2)
library(cowplot)
```

```

Attaching package: 'cowplot'

The following object is masked from 'package:lubridate':

    stamp
```

```r
library(specr)
library(dplyr)
library(tidyr)
library(patchwork)
```

```

Attaching package: 'patchwork'

The following object is masked from 'package:cowplot':

    align_plots
```

```r
library(kSamples)
```

```
Loading required package: SuppDists
```

```r
# Load data
df_small <- readRDS(here("data/results/all_models_final_results_1920_2026-04-15.RDS"))
df_large <- readRDS(here("data/results/all_models_final_results_Univ100_6400_2026-04-20.RDS")
```

NAs

```r
# Replace Inf and 0.000 by NA
# as both indicate model non-convergence
df_large <- df_large %>%
  mutate(RR = ifelse(model_type == "GEE" & (RR == Inf | RR == 0), NA, RR),
         conf.low = ifelse(model_type == "GEE" & (conf.low == Inf | conf.low == 0), NA, conf
         conf.high = ifelse(model_type == "GEE" & (conf.high == Inf | conf.high == 0), NA, c

# How many NAs?
df_large %>%
  summarise(na_count = sum(is.na(RR)))
```

```
# A tibble: 1 x 1
  na_count
     <int>
1     1487
```

```r
# How are NAs distributed per decision?
df_large %>%
  group_by(model_type) %>%
  summarise(na_count = sum(is.na(RR)))
```

```
# A tibble: 6 x 2
  model_type  na_count
  <chr>          <int>
1 GEE              221
2 cox_haz            0
3 log-binom       1102
4 log_reg          164
5 mix_log_reg        0
6 time_mix           0
```

```r
df_large %>%
  group_by(data_type) %>%
  summarise(na_count = sum(is.na(RR)))
```

```
# A tibble: 4 x 2
  data_type     na_count
  <chr>            <int>
1 cross_all          403
2 cross_w1           110
3 long_nostrict      411
4 long_w1            563
```

```r
df_large %>%
  group_by(outcome_label) %>%
  summarise(na_count = sum(is.na(RR)))
```

```
# A tibble: 7 x 2
  outcome_label                                             na_count
  <chr>                                                        <int>
1 age_at_event+death_by_event+event_ever                         177
2 age_at_event+death_by_event+event_ever+since_last_int          193
3 age_at_event+event_ever                                        266
4 age_at_event+event_ever+since_last_int                         216
5 death_by_event+event_ever                                      187
6 death_by_event+event_ever+since_last_int                       182
7 event_ever                                                     266
```

```r
df_large %>%
  group_by(exclusion) %>%
  summarise(na_count = sum(is.na(RR)))
```

```
# A tibble: 6 x 2
  exclusion          na_count
  <chr>                 <int>
1 born_after1954          209
2 country                 261
3 divorced_widowed        277
4 married_before18        248
5 none                    263
6 younger_than_55_w1      229
```

```r
df_large %>%
  group_by(missing_data) %>%
  summarise(na_count = sum(is.na(RR)))
```

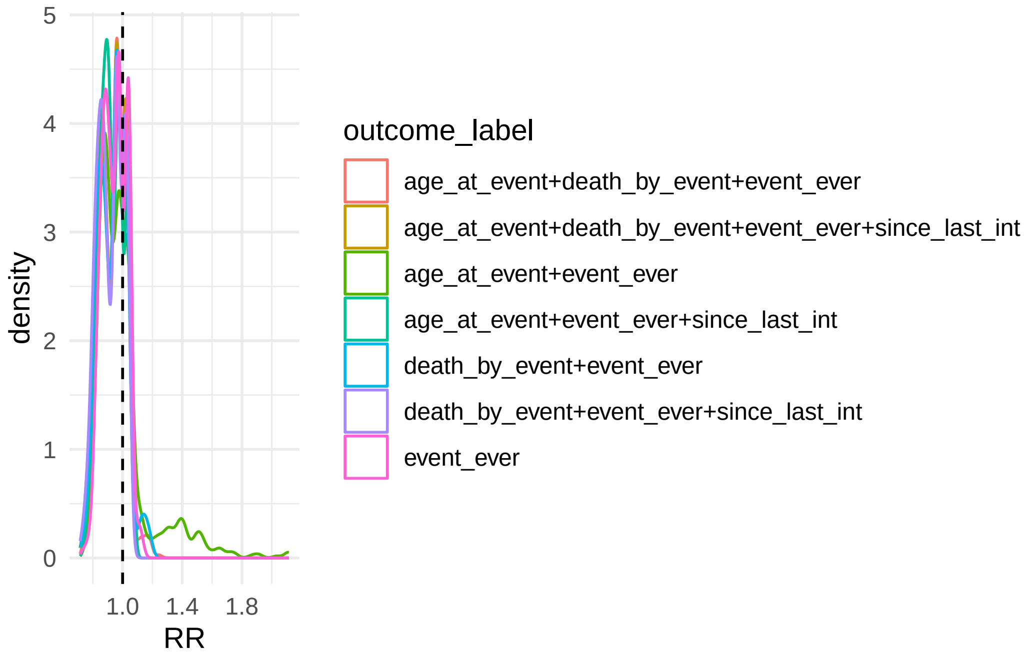
```
# A tibble: 2 x 2
  missing_data na_count
  <chr>           <int>
1 cc                637
2 ni                850
```

OUTCOME PLOT

```r
#### OUTCOME ####
OUT <- ggplot(df_large, aes(x = RR, colour = outcome_label)) +
  geom_density() + theme_minimal() + geom_vline(xintercept = 1, linetype = "dashed")  #+ ther

OUT
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

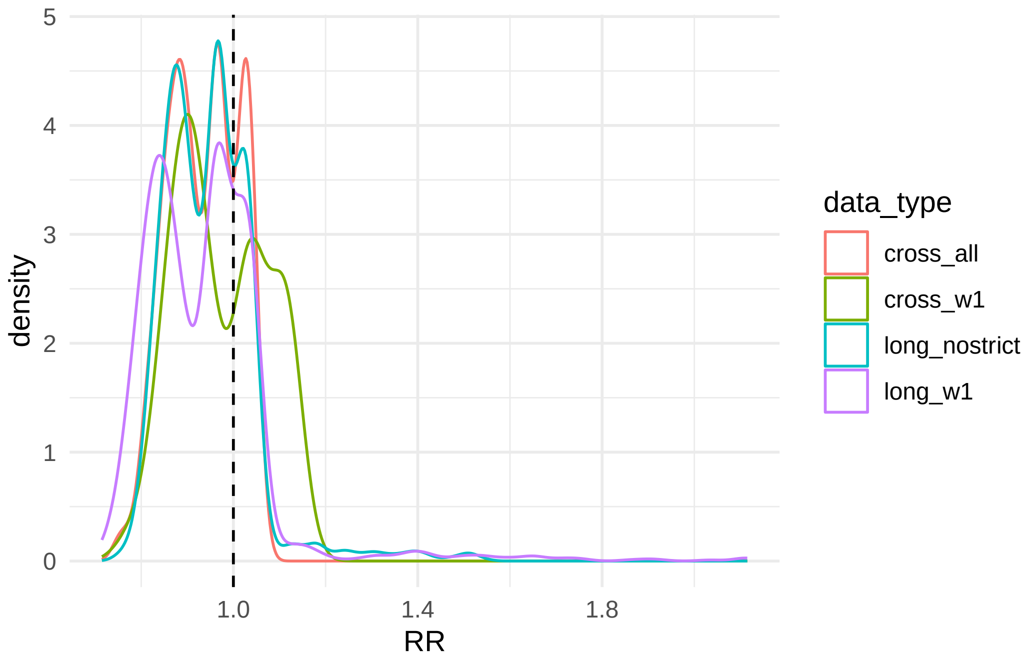


DATA TYPE PLOT

```
#### DATA TYPE ####
DT <- ggplot(df_large, aes(x = RR, colour = data_type)) +
  geom_density() + theme_minimal() + geom_vline(xintercept = 1, linetype = "dashed")  #+ ther

DT
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

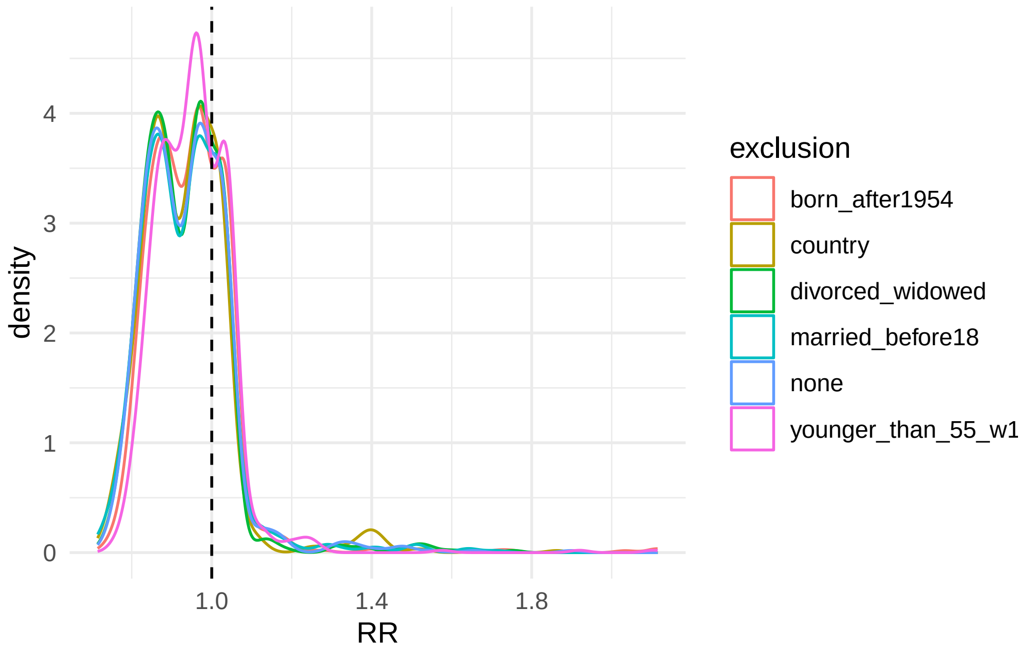


EXCLUSION PLOT

```
#### EXCLUSION ####
EC <- ggplot(df_large, aes(x = RR, colour = exclusion)) +
  geom_density() + theme_minimal() + geom_vline(xintercept = 1, linetype = "dashed")  #+ them

EC
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

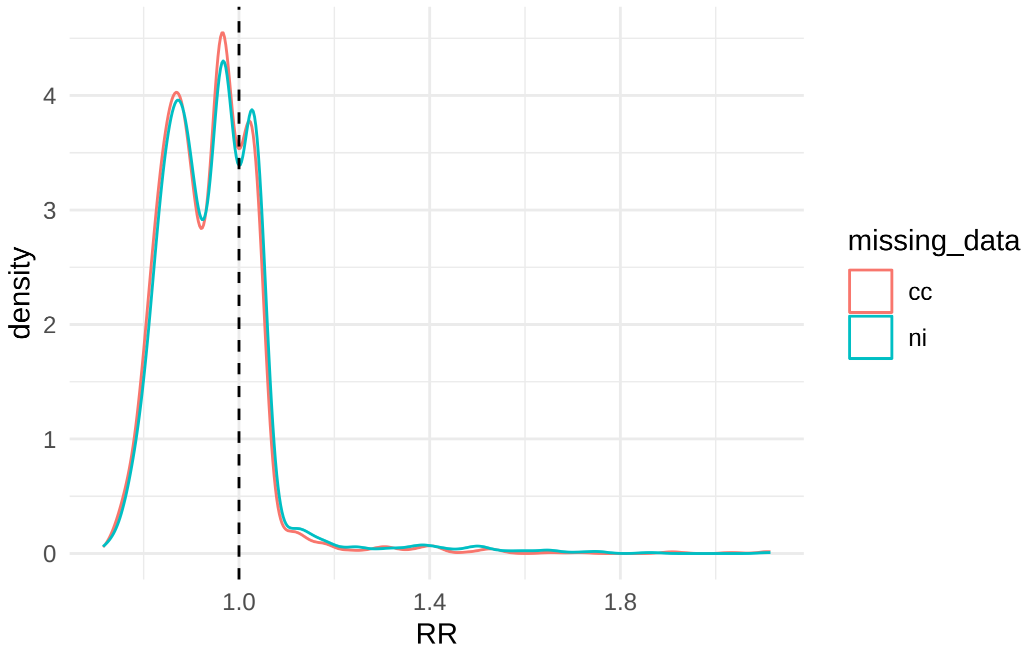


MISSING DATA PLOT

```
#### MISSING DATA ####
MD <- ggplot(df_large, aes(x = RR, colour = missing_data)) +
  geom_density() + theme_minimal()  + geom_vline(xintercept = 1, linetype = "dashed") #+ ther

MD
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

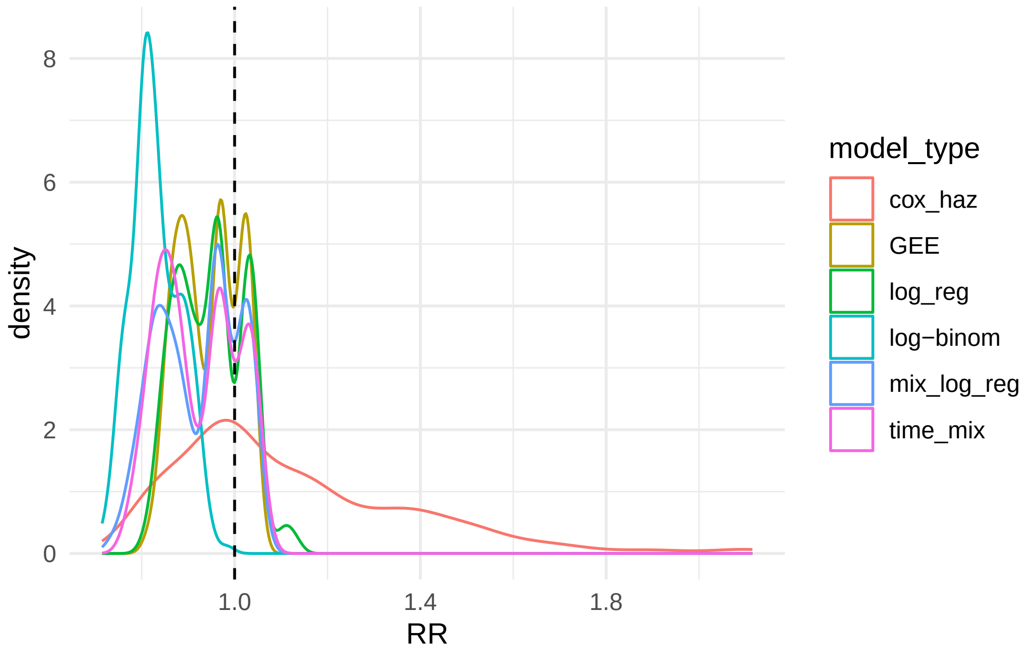


MODEL TYPE PLOT

```
#### MODEL TYPE ####
SM <- ggplot(df_large, aes(x = RR, colour = model_type)) +
  geom_density() + theme_minimal() + geom_vline(xintercept = 1, linetype = "dashed")
 #+ theme(legend.position="none")

SM
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

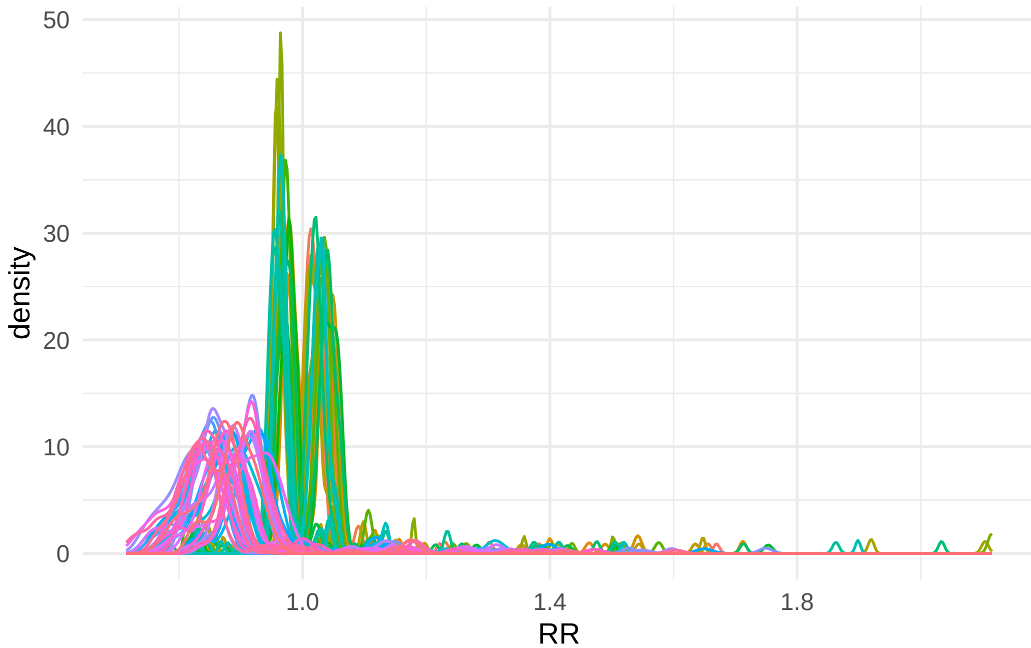


ADJUSTMENT LABEL PLOT

```
#### ADJUSTMENT LABEL ####
ADJ <- ggplot(df_large, aes(x = RR, colour = adjustment_label)) +
  geom_density() + theme_minimal() + theme(legend.position="none")

ADJ
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

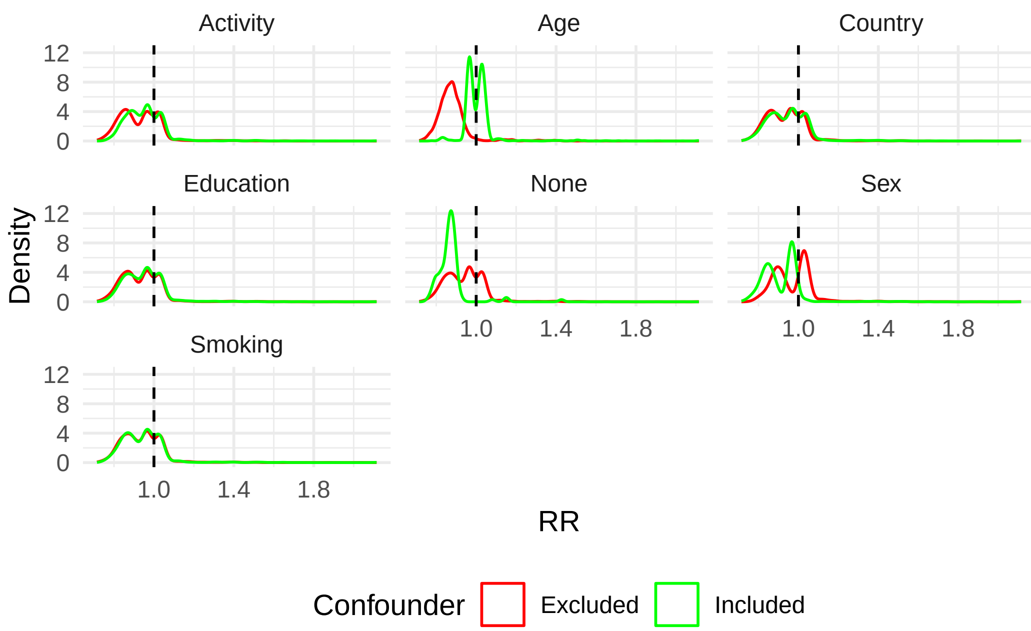



```
# Create binary indicators for each confounder
df_adj <- df_large %>%
  mutate(
    has_none = grepl("none", adjustment_label, ignore.case = TRUE),
    has_age = grepl("age_int", adjustment_label, ignore.case = TRUE),
    has_sex = grepl("dn042_", adjustment_label, ignore.case = TRUE),
    has_country = grepl("country", adjustment_label, ignore.case = TRUE),
    has_smoking = grepl("br001_", adjustment_label, ignore.case = TRUE),
    has_education = grepl("dn010_", adjustment_label, ignore.case = TRUE),
    has_activity = grepl("phactiv", adjustment_label, ignore.case = TRUE)
  )

# Plot density for each confounder
df_adj %>%
  pivot_longer(
    cols = starts_with("has_"),
    names_to = "confounder",
    values_to = "included"
  ) %>%
  mutate(confounder = gsub("has_", "", confounder) %>% str_to_title()) %>%
  ggplot(aes(x = RR, colour = included)) +
  geom_density(alpha = 0.5) +
  geom_vline(xintercept = 1, linetype = "dashed") +
  facet_wrap(~ confounder, ncol = 3) +
```

```
  scale_colour_manual(
    name = "Confounder",
    values = c("TRUE" = "green", "FALSE" = "red"),
    labels = c("TRUE" = "Included", "FALSE" = "Excluded")
  ) +
  labs(x = "RR", y = "Density") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
Warning: Removed 10409 rows containing non-finite outside the scale range
(`stat_density()`).
```



GRID PLOT

```
plot_grid(DT, EC, MD, OUT, ADJ, SM, labels = c("DT", "EC", "MD", "OUT", "ADJ", "SM"), cols =
```

```
Warning in plot_grid(DT, EC, MD, OUT, ADJ, SM, labels = c("DT", "EC", "MD", :
Argument 'cols' is deprecated. Use 'ncol' instead.
```

```
Warning: Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```

```
Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
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(`stat_density()`).
Removed 1487 rows containing non-finite outside the scale range
(`stat_density()`).
```
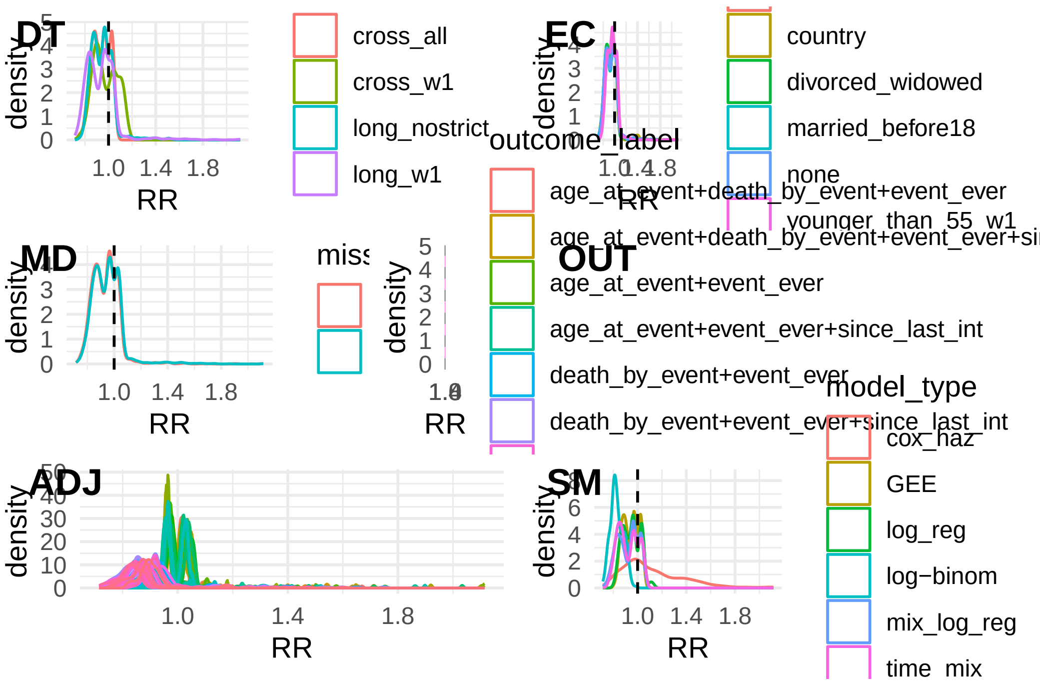
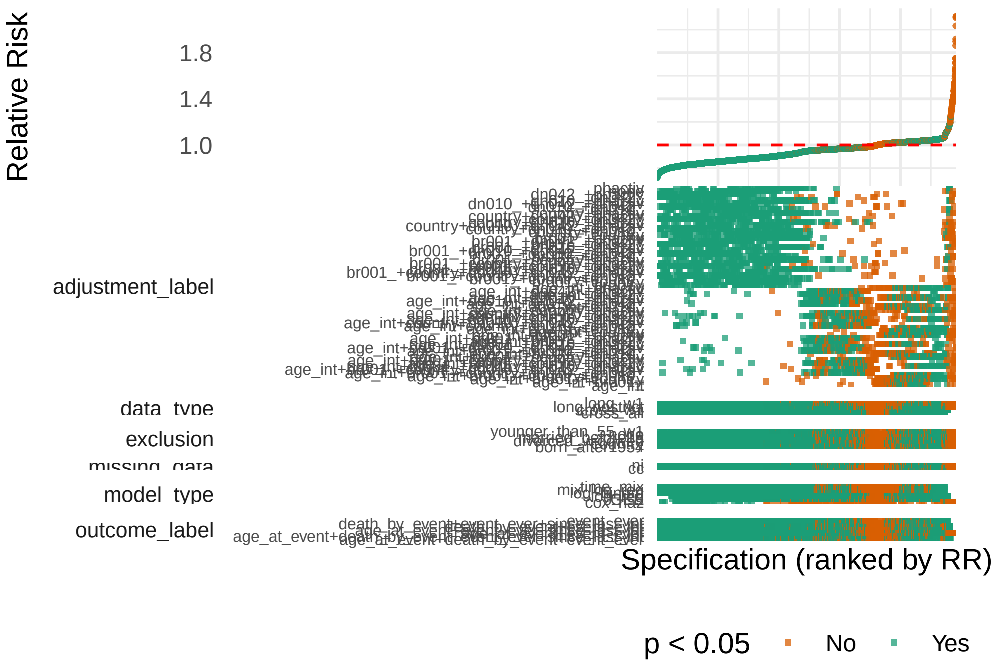



SPECIFICATION CURVE - SIGNIFICANCE

```
df_plot <- df_large %>%
  filter(!is.na(RR)) |>
  arrange(RR) %>%
  mutate(spec_id = row_number(),
         significant = p.value < 0.05
)

# Top: estimates
p_top <- ggplot(df_plot, aes(x = spec_id, y = RR, colour = significant)) +
```

```r
  geom_point(size = 0.5, alpha = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "red") +
  scale_colour_manual(
    values = c("TRUE" = "#1b9e77", "FALSE" = "#d95f02"),
    guide = "none"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    plot.margin = margin(5, 5, 0, 5)
  ) +
  labs(y = "Relative Risk", x = NULL)

# Bottom: specification grid
p_bottom <- df_plot %>%
  select(spec_id, data_type, missing_data, exclusion, outcome_label, model_type, adjustment_l
  pivot_longer(
    cols = c(data_type, missing_data, exclusion, outcome_label, model_type, adjustment_label)
    names_to = "choice",
    values_to = "value"
  ) %>%
  ggplot(aes(x = spec_id, y = value, colour = significant)) +
  geom_point(shape = 15, size = 0.8, alpha = 0.75) +
  scale_colour_manual(
    name = "p < 0.05",
    values = c("TRUE" = "#1b9e77", "FALSE" = "#d95f02"),
    labels = c("TRUE" = "Yes", "FALSE" = "No")
  ) +
  facet_grid(choice ~ ., scales = "free_y", space = "free_y", switch = "y") +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(),
    strip.placement = "outside",
    strip.text.y.left = element_text(angle = 0, hjust = 1, size = 8),
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 6)
  ) +
  labs(x = "Specification (ranked by RR)", y = NULL)

p_top <- p_top + scale_x_continuous(expand = c(0, 0))
```

```
p_bottom <- p_bottom + scale_x_continuous(expand = c(0, 0))


(p_top / p_bottom) +
  plot_layout(heights = c(1, 2)) &
  theme(plot.margin = margin(0, 10, 0, 10))
```



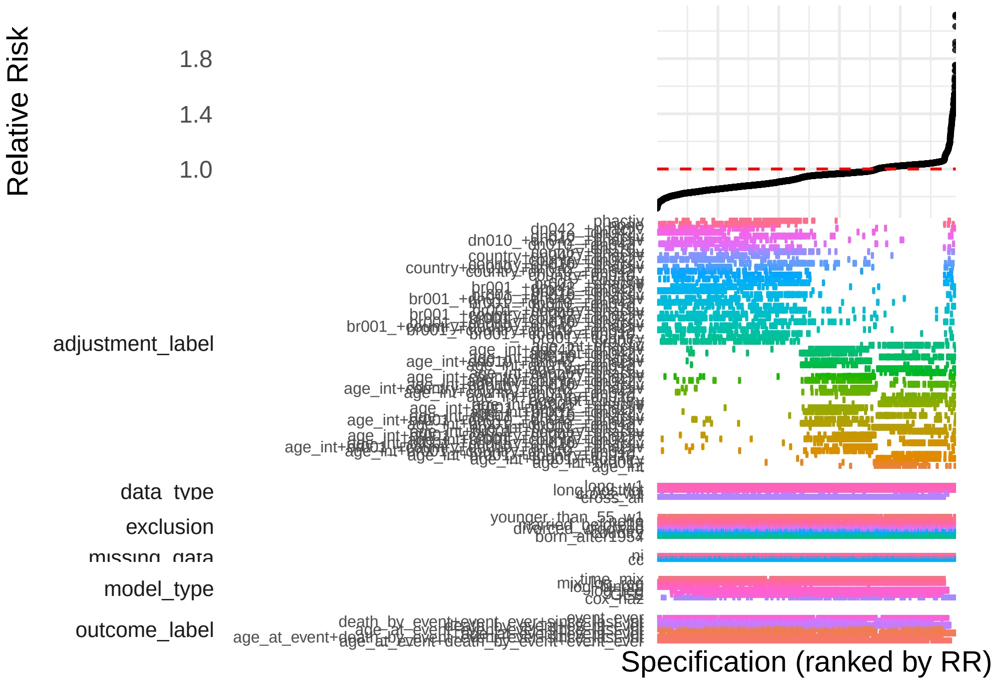
SPECIFICATION CURVE

```
df_plot <- df_large %>%
  filter(!is.na(RR)) |>
  arrange(RR) %>%
  mutate(spec_id = row_number())

# Top: estimates
p_top <- ggplot(df_plot, aes(x = spec_id, y = RR)) +
  geom_point(size = 0.5, alpha = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "red") +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
```

```
    plot.margin = margin(5, 5, 0, 5)
  ) +
  labs(y = "Relative Risk", x = NULL)

# Bottom: specification grid
p_bottom <- df_plot %>%
  select(spec_id, data_type, missing_data, exclusion, outcome_label, model_type, adjustment_l
  pivot_longer(
    cols = c(data_type, missing_data, exclusion, outcome_label, model_type, adjustment_label
    names_to = "choice",
    values_to = "value"
  ) %>%
  ggplot(aes(x = spec_id, y = value, colour = value)) +
  geom_tile(linewidth = 0.5, size = 0.8) +
  facet_grid(choice ~ ., scales = "free_y", space = "free_y", switch = "y") +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(),
    strip.placement = "outside",
    strip.text.y.left = element_text(angle = 0, hjust = 1, size = 8),
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 6)
  ) + scale_fill_brewer(palette = "Paired")  +
  labs(x = "Specification (ranked by RR)", y = NULL) + theme(legend.position="none")
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

```
p_top <- p_top + scale_x_continuous(expand = c(0, 0))
p_bottom <- p_bottom + scale_x_continuous(expand = c(0, 0))

(p_top / p_bottom) +
  plot_layout(heights = c(1, 2)) &
  theme(plot.margin = margin(0, 10, 0, 10))
```




```
largeRRs <- df_large |>
  mutate(large = ifelse(RR > 1.1, 1, 0))

sum(largeRRs$large, na.rm=T)
```

```
[1] 171
```

```
# Large sample
#ad.test(RR ~ missing_data , data = df_plot)
#ad.test(RR ~ data_type , data = df_plot)
#ad.test(RR ~ exclusion , data = df_plot)
#ad.test(RR ~ outcome_label, data = df_plot)
#ad.test(RR ~ adjustment_label , data = df_plot)
#ad.test(RR ~ model_type, data = df_plot)

df_small <- df_small %>%
  mutate(RR = ifelse(model_type == "GEE" & (RR == Inf | RR == 0), NA, RR),
         conf.low = ifelse(model_type == "GEE" & (conf.low == Inf | conf.low == 0), NA, conf
         conf.high = ifelse(model_type == "GEE" & (conf.high == Inf | conf.high == 0), NA, co

df_small_noNAs <- df_small |>
  filter(!is.na(RR))
```

```
ad.test(RR ~ missing_data , data = df_small_noNAs)
```

```
 Anderson-Darling k-sample test.

Number of samples:  2
Sample sizes:  759, 698
Number of ties: 19

Mean of  Anderson-Darling  Criterion: 1
Standard deviation of  Anderson-Darling  Criterion: 0.76039

T.AD = ( Anderson-Darling  Criterion - mean)/sigma

Null Hypothesis: All samples come from a common population.

               AD    T.AD  asympt. P-value
version 1: 1.2455 0.32283          0.25055
version 2: 1.2500 0.32459          0.25017
```

```
ad.test(RR ~ data_type , data = df_small_noNAs)
```

```
 Anderson-Darling k-sample test.

Number of samples:  4
Sample sizes:  367, 41, 537, 512
Number of ties: 19

Mean of  Anderson-Darling  Criterion: 3
Standard deviation of  Anderson-Darling  Criterion: 1.31707

T.AD = ( Anderson-Darling  Criterion - mean)/sigma

Null Hypothesis: All samples come from a common population.

               AD   T.AD  asympt. P-value
version 1: 30.816 21.120       8.4826e-14
version 2: 30.900 21.151       7.3014e-14
```

```r
ad.test(RR ~ exclusion , data = df_small_noNAs)
```

```
 Anderson-Darling k-sample test.

Number of samples:  6
Sample sizes:  247, 256, 242, 228, 233, 251
Number of ties: 19

Mean of  Anderson-Darling  Criterion: 5
Standard deviation of  Anderson-Darling  Criterion: 1.69795

T.AD = ( Anderson-Darling  Criterion - mean)/sigma

Null Hypothesis: All samples come from a common population.

               AD   T.AD  asympt. P-value
version 1: 15.036 5.9107       0.00015079
version 2: 15.000 5.9148       0.00015024
```

```r
ad.test(RR ~ outcome_label, data = df_small_noNAs)
```

```
 Anderson-Darling k-sample test.

Number of samples:  7
Sample sizes:  228, 178, 230, 169, 232, 191, 229
Number of ties: 19

Mean of  Anderson-Darling  Criterion: 6
Standard deviation of  Anderson-Darling  Criterion: 1.85939

T.AD = ( Anderson-Darling  Criterion - mean)/sigma

Null Hypothesis: All samples come from a common population.

               AD   T.AD  asympt. P-value
version 1: 33.027 14.535       1.6066e-11
version 2: 33.100 14.564       1.4860e-11
```

```r
ad.test(RR ~ adjustment_label , data = df_small_noNAs)
```

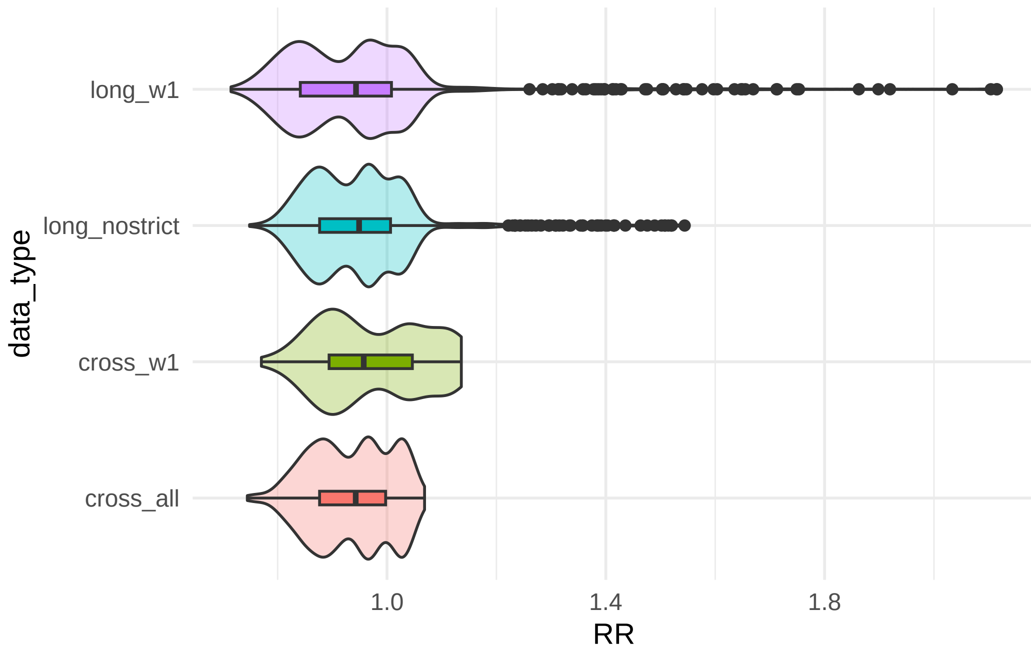
```
 Anderson-Darling k-sample test.

Number of samples:  64
Sample sizes:  22, 25, 21, 23, 26, 19, 20, 22, 22, 16, 21, 20, 18, 25, 20, 21, 25, 26, 17, 19
Number of ties: 19

Mean of  Anderson-Darling  Criterion: 63
Standard deviation of  Anderson-Darling  Criterion: 5.90617

T.AD = ( Anderson-Darling  Criterion - mean)/sigma

Null Hypothesis: All samples come from a common population.

               AD   T.AD  asympt. P-value
version 1: 709.62 109.48      1.3361e-165
version 2: 710.00 109.52      1.0653e-165
```

```r
ad.test(RR ~ model_type, data = df_small_noNAs)
```

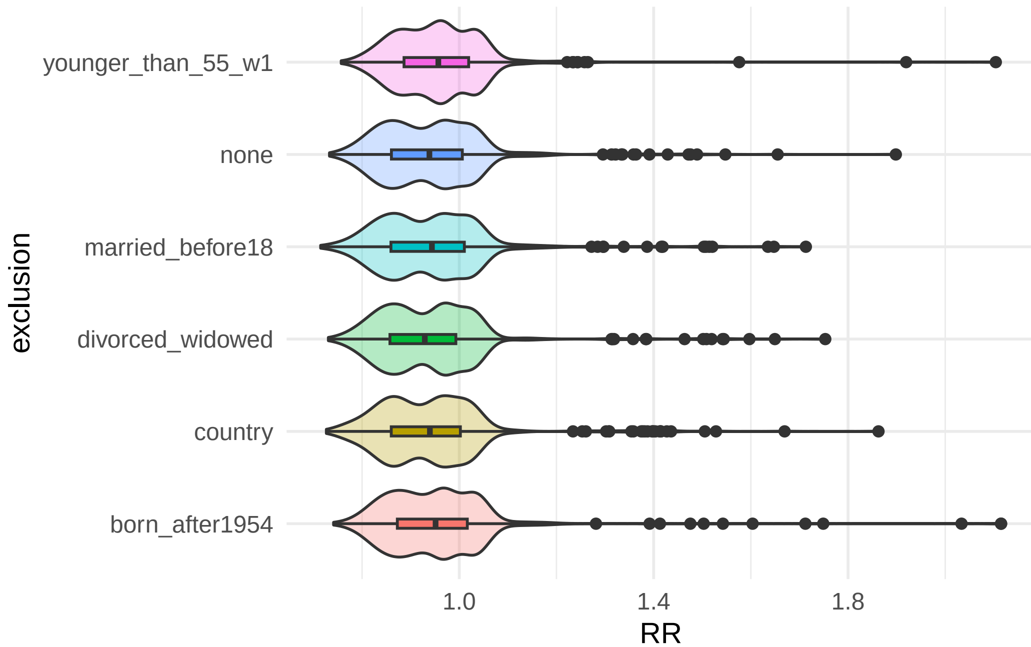
```
 Anderson-Darling k-sample test.

Number of samples:  6
Sample sizes:  103, 339, 356, 55, 365, 239
Number of ties: 19

Mean of  Anderson-Darling  Criterion: 5
Standard deviation of  Anderson-Darling  Criterion: 1.69853

T.AD = ( Anderson-Darling  Criterion - mean)/sigma

Null Hypothesis: All samples come from a common population.

               AD   T.AD  asympt. P-value
version 1: 157.01 89.493       3.1610e-67
version 2: 157.00 89.658       1.6053e-67
```

```r
# Violin plots outcome
OUT_vio <- ggplot(df_plot, aes(x = outcome_label, y = RR, fill = outcome_label)) +
  geom_violin(alpha=0.3) +
  geom_boxplot(width = 0.1) +
  coord_flip() + theme_minimal() + theme(legend.position = "none")

# Violin plots data_type
DT_vio <- ggplot(df_plot, aes(x = data_type, y = RR, fill = data_type)) +
  geom_violin(alpha=0.3) +
  geom_boxplot(width = 0.1) +
  coord_flip() + theme_minimal() + theme(legend.position = "none")

# Violin plots missing_data
MD_vio <- ggplot(df_plot, aes(x = missing_data, y = RR, fill = missing_data)) +
  geom_violin(alpha=0.3) +
  geom_boxplot(width = 0.1) +
  coord_flip() + theme_minimal() + theme(legend.position = "none")

# Violin plots exclusion
EC_vio <- ggplot(df_plot, aes(x = exclusion, y = RR, fill = exclusion)) +
  geom_violin(alpha=0.3) +
  geom_boxplot(width = 0.1) +
  coord_flip() + theme_minimal() + theme(legend.position = "none")

# Violin plots adjustment_label all
#ggplot(df_plot, aes(x = adjustment_label, y = RR, fill = adjustment_label)) +
#  geom_violin(alpha=0.3) +
#  geom_boxplot(width = 0.1) +
#  coord_flip() + theme_minimal() + theme(legend.position = "none")

# Violin plots adjustment_label per confounder (in/excluded)
ADJ_vio <- df_adj %>%
  pivot_longer(
    cols = starts_with("has_"),
    names_to = "confounder",
    values_to = "included"
  ) %>%
  mutate(confounder = gsub("has_", "", confounder) %>% str_to_title()) %>%
  ggplot(aes(x = confounder, y = RR, fill = included)) +
  geom_violin(alpha = 0.3, position = position_dodge(0.9)) +
  geom_boxplot(width = 0.1, position = position_dodge(0.9)) +
  coord_flip() +
```

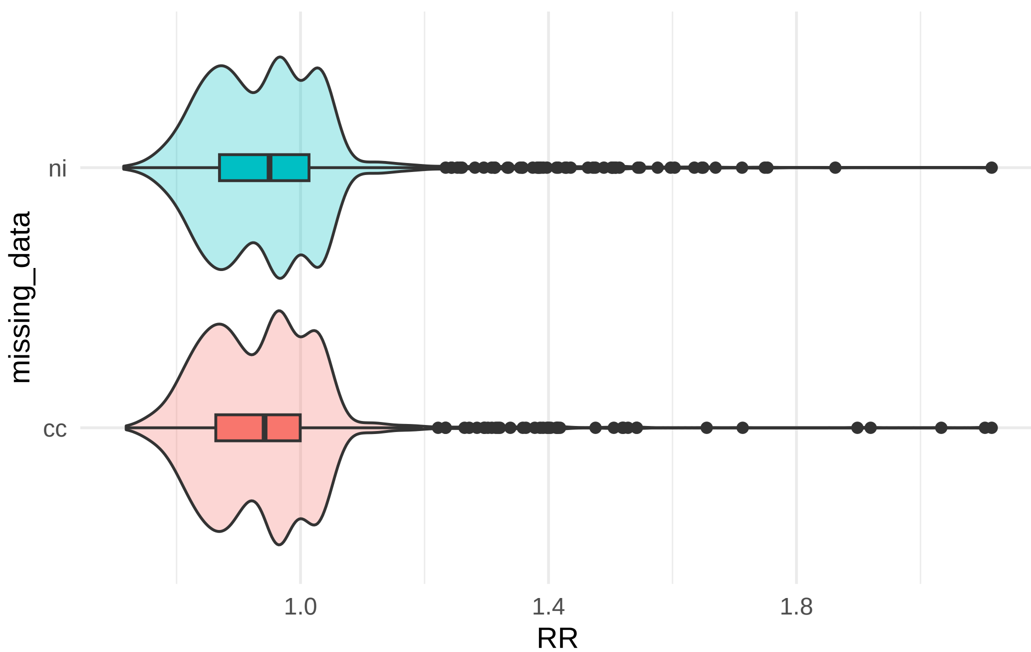
```r
  scale_fill_manual(
    values = c("TRUE" = "#1b9e77", "FALSE" = "#d95f02"),
    labels = c("TRUE" = "Included", "FALSE" = "Excluded")
  ) +
  theme_minimal() +
  labs(fill = "Confounder") + theme(legend.position = "none")


# Violin plots model_type
SM_vio <- ggplot(df_plot, aes(x = model_type, y = RR, fill = model_type)) +
  geom_violin(alpha=0.3) +
  geom_boxplot(width = 0.1) +
  coord_flip() + theme_minimal() + theme(legend.position = "none")

DT_vio
```



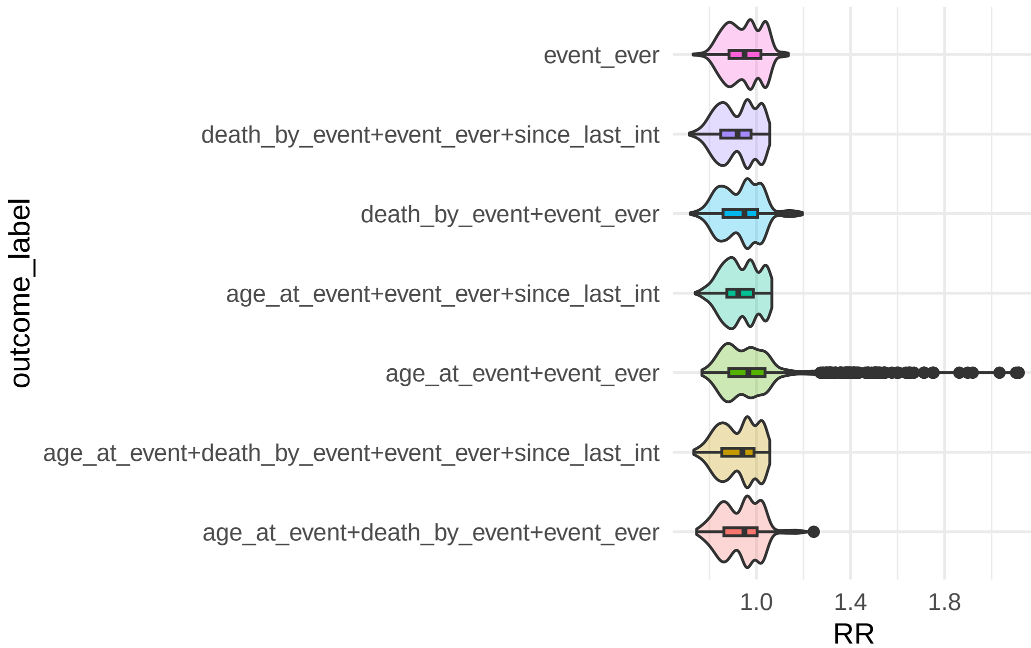

```r
EC_vio
```




```
MD_vio
```



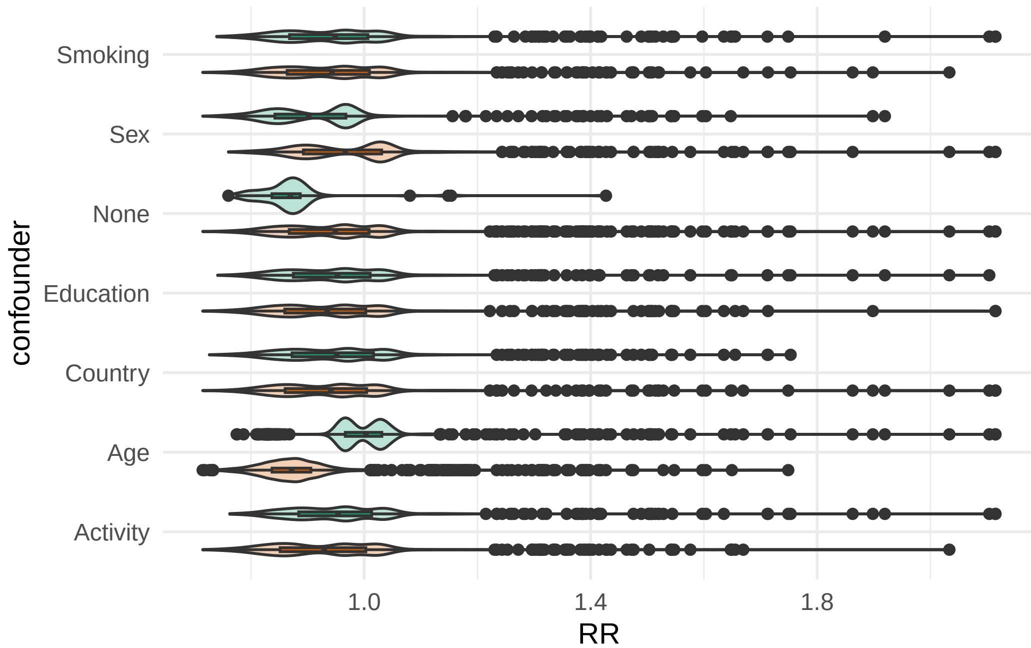

```
OUT_vio
```



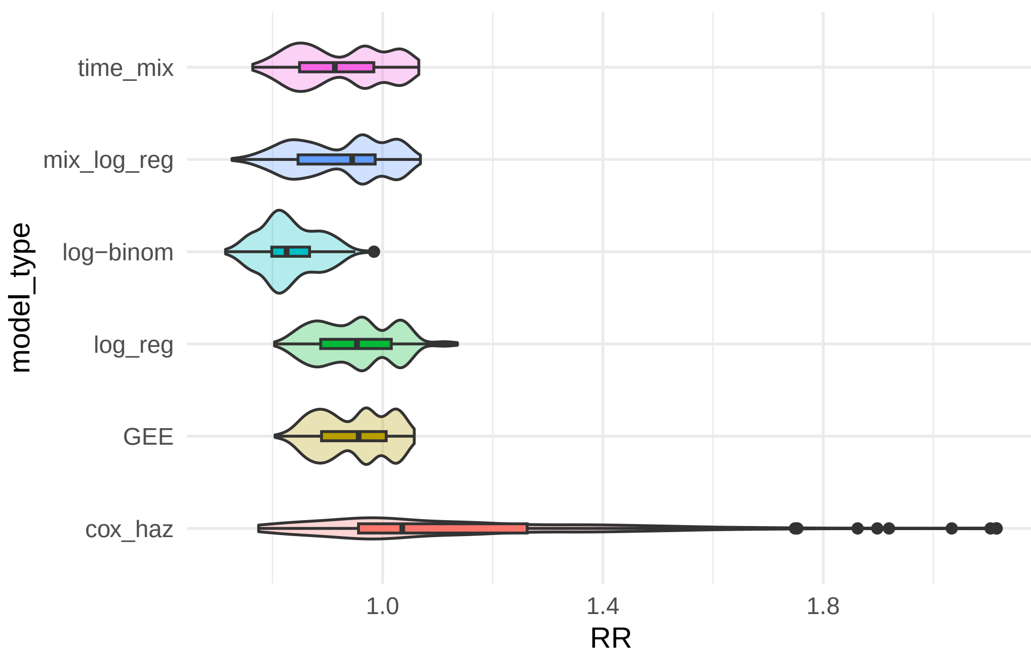

```
ADJ_vio
```

```
Warning: Removed 10409 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

```
Warning: Removed 10409 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```


Smoking
Sex
None
Education
Country
Age
Activity
confounder
1.0
1.4
1.8
RR

```
SM_vio
```


time_mix
mix_log_reg
log−binom
log_reg
GEE
cox_haz
model_type
1.0
1.4
1.8
RR

SPECIFICATION CURVE CLEANER WITH CI

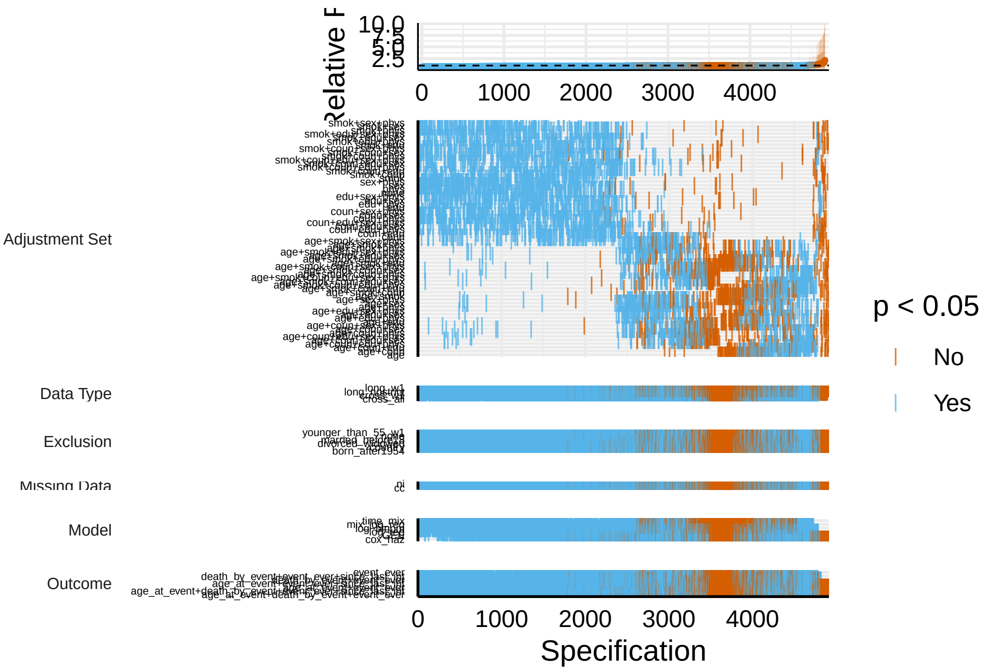
```r
library(ggplot2)
library(dplyr)
library(tidyr)
library(cowplot)
library(stringr)

df_plot <- df_large %>%
  filter(!is.na(RR)) %>%
  arrange(RR) %>%
  mutate(
    spec_id = row_number(),
    significant = p.value < 0.05,
    adjustment_label = adjustment_label %>%
      str_replace_all("dn010_", "edu") %>%
      str_replace_all("dn042_", "sex") %>%
      str_replace_all("br001_", "smok") %>%
      str_replace_all("phactiv", "phys") %>%
      str_replace_all("country", "coun") %>%
      str_replace_all("age_int", "age") %>%
      str_replace_all("none", "none")
  )

choice_labels <- c(
  "data_type" = "Data Type",
  "missing_data" = "Missing Data",
  "exclusion" = "Exclusion",
  "outcome_label" = "Outcome",
  "model_type" = "Model",
  "adjustment_label" = "Adjustment Set"
)

# Set common x limits
x_limits <- c(0, max(df_plot$spec_id))

# Top panel: estimates with CIs
p_top <- ggplot(df_plot, aes(x = spec_id, y = RR, colour = significant)) +
  geom_linerange(
    aes(ymin = conf.low, ymax = conf.high),
    linewidth = 0.1, alpha = 0.1
  ) +
```

```r
  geom_point(aes(color = significant),
             size = 0.5) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "black", linewidth = 0.3) +
  scale_colour_manual(
    values = c("TRUE" = "#56B4E9", "FALSE" = "#D55E00"),
    guide = "none"
  ) +
  scale_y_continuous(expand = c(0.02, 0)) +
  scale_x_continuous(expand = c(0.01, 0)) +
  theme_minimal()+
  theme(strip.text = element_blank(),
        axis.line = element_line("black", size = .3),
        legend.position = "none",
        panel.spacing = unit(1, "lines"),
        axis.text = element_text(colour = "black")) +
  labs(y = "Relative Risk", x = NULL)
```

```
Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
i Please use the `linewidth` argument instead.
```

```r
# Bottom panel: specification grid
p_bottom <- df_plot %>%
  select(spec_id, data_type, missing_data, exclusion, outcome_label, model_type, adjustment_l
  pivot_longer(
    cols = c(data_type, missing_data, exclusion, outcome_label, model_type, adjustment_label)
    names_to = "choice",
    values_to = "value"
  ) %>%
  ggplot(aes(x = spec_id, y = value, colour = significant)) +
  geom_point(shape = 124,
             size = 2, alpha = 0.8) +
  scale_colour_manual(
    name = "p < 0.05",
    values = c("TRUE" = "#56B4E9", "FALSE" = "#D55E00"),
    labels = c("TRUE" = "Yes", "FALSE" = "No")
  ) +
  scale_x_continuous(limits = x_limits, expand = c(0, 0)) +
  facet_grid(choice ~ ., scales = "free_y", space = "free_y", switch = "y",
             labeller = labeller(choice = choice_labels)) +
  theme_minimal() +
  theme(
```

```r
    axis.line = element_line("black", size = .5),
    panel.spacing = unit(.75, "lines"),
    axis.text = element_text(colour = "black"),
    axis.text.y = element_text(size = 4),
    strip.text.y.left = element_text(angle = 0, hjust = 1, size = 6),
    strip.placement = "outside",
    plot.margin = margin(0, 10, 5, 10)
  ) +
  labs(x = "Specification", y = NULL)

# Combine
plot_grid(p_top, p_bottom, ncol = 1, align = "v", axis = "lr", rel_heights = c(1, 5))
```



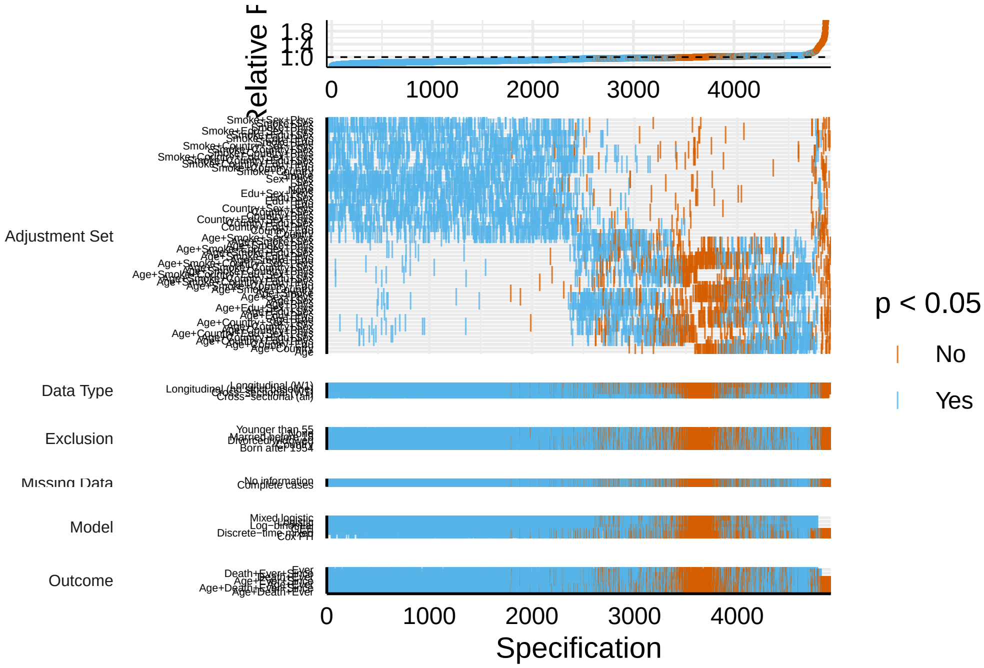
SPECIFICATION CURVE CLEANER WITHOUT CI

```r
library(ggplot2)
library(dplyr)
library(tidyr)
library(cowplot)
library(stringr)
```

```
df_plot <- df_large %>%
  filter(!is.na(RR)) %>%
  arrange(RR) %>%
  mutate(
    spec_id = row_number(),
    significant = p.value < 0.05,

    # Adjustment labels
    adjustment_label = adjustment_label %>%
      str_replace_all("dn010_", "Edu") %>%
      str_replace_all("dn042_", "Sex") %>%
      str_replace_all("br001_", "Smoke") %>%
      str_replace_all("phactiv", "Phys") %>%
      str_replace_all("country", "Country") %>%
      str_replace_all("age_int", "Age") %>%
      str_replace_all("none", "None"),

    # Outcome labels
    outcome_label = outcome_label %>%
      str_replace_all("age_at_event", "Age") %>%
      str_replace_all("event_ever", "Ever") %>%
      str_replace_all("death_by_event", "Death") %>%
      str_replace_all("since_last_int", "Since"),

    # Data type labels
    data_type = data_type %>%
      str_replace_all("cross_all", "Cross-sectional (all)") %>%
      str_replace_all("cross_w1", "Cross-sectional (W1)") %>%
      str_replace_all("long_nostrict", "Longitudinal (no strict baseline)") %>%
      str_replace_all("long_w1", "Longitudinal (W1)"),

    # Missing data labels
    missing_data = missing_data %>%
      str_replace_all("cc", "Complete cases") %>%
      str_replace_all("ni", "No information"),

    # Exclusion labels
    exclusion = exclusion %>%
      str_replace_all("born_after1954", "Born after 1954") %>%
      str_replace_all("younger_than_55_w1", "Younger than 55") %>%
      str_replace_all("married_before18", "Married before 18") %>%
      str_replace_all("divorced_widowed", "Divorced/widowed") %>%
```

```
      str_replace_all("country", "Country") %>%
      str_replace_all("none", "None"),

    # Model type labels
    model_type = model_type %>%
      str_replace_all("cox_haz", "Cox PH") %>%
      str_replace_all("mix_log_reg", "Mixed logistic") %>%
      str_replace_all("log_reg", "Logistic") %>%
      str_replace_all("log-binom", "Log-binomial") %>%
      str_replace_all("time_mix", "Discrete-time mixed") %>%
      str_replace_all("GEE", "GEE")
  )

choice_labels <- c(
  "data_type" = "Data Type",
  "missing_data" = "Missing Data",
  "exclusion" = "Exclusion",
  "outcome_label" = "Outcome",
  "model_type" = "Model",
  "adjustment_label" = "Adjustment Set"
)

# Set common x limits
x_limits <- c(0, max(df_plot$spec_id))

# Top panel: estimates with CIs
p_top <- ggplot(df_plot, aes(x = spec_id, y = RR, colour = significant)) +
  geom_point(aes(color = significant),
             size = 0.5) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "black", linewidth = 0.3) +
  scale_colour_manual(
    values = c("TRUE" = "#56B4E9", "FALSE" = "#D55E00"),
    guide = "none"
  ) +
  scale_y_continuous(expand = c(0.02, 0)) +
  scale_x_continuous(expand = c(0.01, 0)) +
  theme_minimal()+
  theme(strip.text = element_blank(),
        axis.line = element_line("black", size = .3),
        legend.position = "none",
        panel.spacing = unit(1, "lines"),
        axis.text = element_text(colour = "black")) +
```

```r
  labs(y = "Relative Risk", x = NULL)

# Bottom panel: specification grid
p_bottom <- df_plot %>%
  select(spec_id, data_type, missing_data, exclusion, outcome_label, model_type, adjustment_l
  pivot_longer(
    cols = c(data_type, missing_data, exclusion, outcome_label, model_type, adjustment_label)
    names_to = "choice",
    values_to = "value"
  ) %>%
  ggplot(aes(x = spec_id, y = value, colour = significant)) +
  geom_point(shape = 124,
             size = 2, alpha = 0.8) +
  scale_colour_manual(
    name = "p < 0.05",
    values = c("TRUE" = "#56B4E9", "FALSE" = "#D55E00"),
    labels = c("TRUE" = "Yes", "FALSE" = "No")
  ) +
  scale_x_continuous(limits = x_limits, expand = c(0, 0)) +
  facet_grid(choice ~ ., scales = "free_y", space = "free_y", switch = "y",
           labeller = labeller(choice = choice_labels)) +
  theme_minimal() +
  theme(
    axis.line = element_line("black", size = .5),
    panel.spacing = unit(.75, "lines"),
    axis.text = element_text(colour = "black"),
    axis.text.y = element_text(size = 4),
    strip.text.y.left = element_text(angle = 0, hjust = 1, size = 6),
    strip.placement = "outside",
    plot.margin = margin(0, 10, 5, 10)
  ) +
  labs(x = "Specification", y = NULL)

# Combine
specPlot <- plot_grid(p_top, p_bottom, ncol = 1, align = "v", axis = "lr", rel_heights = c(1
specPlot
```




```r
ggsave(
  "specification_curve.pdf",
  plot = specPlot,
  width = 8.27,
  height = 11.69,
  units = "in"
)
```

```r
# AGE AS A CONFOUNDER?
df <- readRDS(here("data/cleandata/df_all_waves.RDS"))
w1 <- readRDS(here("data/cleandata/w1_clean.RDS"))

source(here("scripts/04_set_up_multiverse/01_data_type_function.R"))
source(here("scripts/04_set_up_multiverse/03_exposure_function.R"))

df_CA <- df_type_function(df,"cross_all", w1)
df_CW1 <- df_type_function(df,"cross_w1", w1)
df_LNS <- df_type_function(df,"long_nostrict", w1)
df_LW1 <- df_type_function(df,"long_w1", w1)

df_CA <- exposure_function(df_CA)
df_CW1 <- exposure_function(df_CW1)
df_LNS <- exposure_function(df_LNS)
```

```r
df_LW1 <- exposure_function(df_LW1)

df_CA |>
  group_by(exposure) |>
  summarise(meanAGE = mean(age_int, na.rm=T),
            DT = "cross_all")
```

```
# A tibble: 2 x 3
  exposure meanAGE DT
     <dbl>   <dbl> <chr>
1        0    69.1 cross_all
2        1    65.0 cross_all
```

```r
df_CW1 |>
  group_by(exposure) |>
  summarise(meanAGE = mean(age_int, na.rm=T),
            DT = "cross_w1")
```

```
# A tibble: 2 x 3
  exposure meanAGE DT
     <dbl>   <dbl> <chr>
1        0    67.7 cross_w1
2        1    62.2 cross_w1
```

```r
df_LNS |>
  group_by(exposure) |>
  summarise(meanAGE = mean(age_int, na.rm=T),
            DT = "long_nostrict")
```

```
# A tibble: 2 x 3
  exposure meanAGE DT
     <dbl>   <dbl> <chr>
1        0    69.3 long_nostrict
2        1    65.3 long_nostrict
```

```r
df_LW1 |>
  group_by(exposure) |>
  summarise(meanAGE = mean(age_int, na.rm=T),
            DT = "long_w1")
```

```
# A tibble: 2 x 3
  exposure meanAGE DT
     <dbl>   <dbl> <chr>
1        0    71.6 long_w1
2        1    66.3 long_w1
```